Integration of User Centered Design and Software Development Process

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Abstract— **User Centered Design is an important aspect of any product to evaluate its success in the market, where there is an abundance of product of similar functionalities. The software products are moving towards the phase where the end user can take the blocks of software, customize and build their own products. This adds more emphasize on the role of UCD in the product development. Many a time, the process for providing a good user experience is treated separate from the process of developing the product or software. There is an increased awareness that the two process need to integrate and proceed hand in hand for success of the product. This paper investigates and proposes the integration model for different development models with the User Centered Design process.**

Keywords— UCD; SDLC; Waterfall model; Iterative Model; Incremental model; Agile Model; Prototype Model

I. INTRODUCTION

With the plethora of the products in the market and user friendliness being one of the differentiating factors among them, there is an increased awareness on the role played by user centered design in product development. There are studies done and literature available on the process and impact of User Centered Design, on choosing the appropriate development model for a given product; however there is a missing link between these two topics. The integration of the development cycle and UCD is not discussed extensively. This paper aims to fill the gap and proposes the integration model for different development models with the User Centered Design process.

In this paper the term product and software are used interchangeably to indicate the deliverable coming out of the software development process. The term SDLC (also indicative of Systems Development Lifecycle Model [2]) is used interchangeable with Software Development methodology in this paper, to indicate the process followed to complete the software development.

II. SOFTWARE DEVELOPMENT PROCESS

Software Development Process or Software Development Lifecycle (SDLC) is a process of grouping the development activities into distinct phases for better planning and management [1]. The basic concept of the SDLC model is always oriented around the core activities - Requirements, Design, Implementation, Testing, Debugging, Deployment and Maintenance. The way these activities are performed – one after the other or in parallel makes them unique.
There are multiple models available for the organizations to choose while developing the software based on the factors of time and resource. The choice is generally a combined decision between the management and the development team, considering multiple factors that may include management, business, technical and project specific needs. Some of the most commonly known models are Waterfall Model [3], Iterative Model [4], Agile Model[5]. Different frameworks like the one suggested in [6] are used to make the choice on the appropriate model for the development. This paper studies the above mentioned model for the integration with UCD.

III. USER CENTERED DESIGN

User Centered Design [7] involves developing the product giving priority the experience that product provides to the user. The aim is to improve the User Interface (UI) design and provide the best user experience. It involves multiple steps - interacting with the end users, identifying their requirements, coming up with the UI prototypes, getting the user response on the prototypes and working on the feedback. The whole process revolves around the needs of the users and how the product can successfully meet these demands.

User Centered Design is not restricted to software but also any product that is marketed. Many hardware products that have made a mark are the results of understanding the user and catering the needs. UCD process involves understanding the user through research activities, analyzing the requirements of the users and coming up with a UI design to meet their expectations. The interface is validated through iterative testing and feedback from the user. It is also essential that the User Experience and Usability is integrated into the software design, development and testing team to ensure that the interface meets the expected standards and requirements to provide the best user experience.

The UCD referred in this paper, is basically designing the look and functionality of the product from user experience (Ux) perspective – including User Experience, Usability, User Interaction. This paper assumes that the development teams are aware and trained to ensure that the committed user experience is included in the course of the development.

IV. Comparative Study of UCD and SDLC

This section explores the integration points available at a generic level to accommodate UCD into the different phases of SDLC.

A. Requirements

This phase involves gathering the requirements for the product. The general approach is to understand the market, identify the area where the proposed feature or product can make a change. Identifying the prospective customers and getting their inputs on the proposed product and summarize the expectation for the next phase of development. The requirement engineering (RE) is a topic of research and discussion by itself. RE has multiple approaches and methodologies to ensure that all aspects – market need, demand, functionality, additional features – of the product get covered.

While the functionality of the product is the main perspective of the requirement, User experience (Ux) that contributes towards UCD is considered as lower priority. This is generally collected through the reviews, surveys and observation of user handling the existing products.

B. Analysis

Analysis is a phase in software lifecycle, which sometimes is integrated with the Requirements stage itself. In this phase, the requirements so far collected are analyzed and split into details. The detailed input helps in the next stage of designing the product. While the previous phase involved in collecting information from the market and users for direct inputs, the analysis help in identifying the indirect or implicit details that are essential for making a successful product.

From the UCD perspective, the process of user research and analysis fits into this phase, where based on the user requirements collected in the previous phase, can be analyzed and the implicit requirements based on the different factors, can be identified.

C. Design

Design stage involves coming up with a design proposal to meet the requirements identified so far. This stage generally involves identifying the technical details and a high level approach for the actual implementation for the software.

In UCD, this stage is basically coming up with a basic prototype. The low fidelity prototyping – where the high level User Interface is identified through sketches, wireframes and story boards in the UCD process can be equated to this stage of SDLC.

D. Development

This is the stage where the actual product is developed or implemented in conformance to the design arrived up earlier. This is a longer phase, where all the requirements are actually brought into life through the software being developed. It is essential that the software development should confirm to the Usability and other user centric requirements that were included into the design. This is the stage where the UI is actually implemented.

From UCD perspective, this stage may not extend to a direct equivalent in UCD process, but can be roughly equated to the high fidelity prototype.

E. Testing

The software testing generally tends to focus on the functionality. The tests are generally derived from the functional requirements. User Centric Design testing involves testing the prototypes with the user and for usability. This ensures that the proposed User Interface prototype meets the expectations of the users.

It is interesting to note that even if the Ux specific needs are included at the requirements stage, there is a tendency to overlook them in the priority list. There is also a general perception that as it should have been validated in the initial stages during requirements prototyping. It is essential to understand that the initial stages of testing would have been on prototype which is not the end product. Testing Ux and Usability on the final product is important to ensure we meet the User Expectation in the final product that goes to the customers. These are generally achieved in many companies through alpha and beta testing process, where some of the prospective customers use a trial version of the product and give their feedback.

While software testing provides the gaps in the committed functionality, the usability and user testing provide input, which can provide the first round feedback from the customer, even before the product is actually released. In case of UCD, this testing is done with the users on the prototype to ensure the required level of user experience is achieved.

F. Evaluation

While the testing of the product provide an first hand evaluation, the development team works on the identified issues trying to fix the gaps in making the product a success. In the list of feedback obtained, the functionality issues take precedence over the user and usability feedback. When aiming for UCD integration, the Ux feedback also needs to be given equal scale as functional gaps.

G. Observations

From the comparison of UCD and SDLC it is observed that though UCD is similar to the SDLC model. The variation between the two is that the iteration between Evaluation, Feedback and Re-designing can be more frequent in UCD than SDLC model to achieve the perfect User Experience. The intent of the comparison is to illustrate that UCD is itself a well-defined process to follow to get best User Experience. UCD is an iterative approach which can be covered as a lifecycle on its own. At first glance this looks like a long and tedious process. This lifecycle is very similar to the software lifecycle and the following table (Table 1) provides a comparison of both.

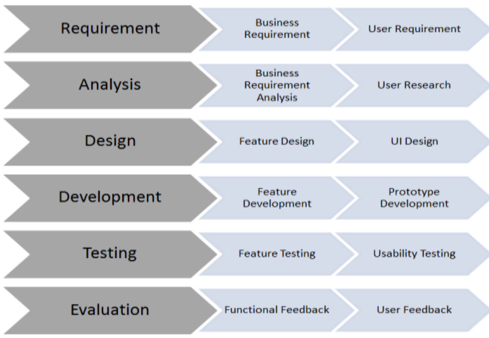

Figure 1 Phases of Software Development

Table 1 Comparison of UCD and SDLC process

Phase	SDLC	UCD
Requirements gathering	Capture Business Requirements	Capture User requirements
Analysis	Detail the Business requirements	User research
Design	Design the feature	Design the UI
Implement	Implement the functionality	Prototype the UI Design
Testing	Test the functionality to match the business requirements	Evaluate the prototype for User requirements through testing
Evaluation	Return the test results for improvement in the next release	The results of Evaluation can go to the Implementation or Design stage based on the results

V. INTEGRATION INTO SPECIFIC SDLC MODELS

The previous section was a generic analysis of how the User Centric Design and Software Development process, can integrate. Hence the approach explained before is only a generic framework which fits while each phase is integrated in a 'as is' model. As the different SDLC models have different approaches for each phase, it is inevitable to customize the integration based on the specific model to get the best results.

This section intends to discuss how UCD fits into different SDLC models. The models considered here are Waterfall Model, Iterative and Agile Model.

A. Waterfall Model

Waterfall model is one of the earliest Software Lifecycle models [3], but it is still used in the industry in different variations and form [8]. This approach requires that the detailing of each phase be clearly documented, expectations be set before hand and subsequent phases work on achieving these set goals. The advantage of this model is the level of details that it can set. The disadvantage is the time period consumed at each stage for detailing.

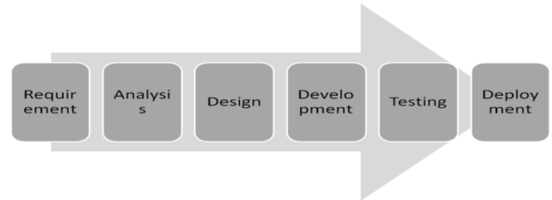

Figure 2 Waterfall Model

UCD can be integrated well in this model, as both involves in detailing. The UCD fits into the Requirement phase of the water fall model. If user requirements can be clubbed with the business requirements, then it becomes easy to get a more accurate design. The complete user experience, the various flows and design of the user interfaces can be designed as a part of the requirement stage and verified with the customer along with the functional requirements.

One of the drawbacks of the waterfall model is that – the requirement gathering sets fixed goals, which by end of the project cycle can become outdated – given the multiple stages in between. The model also does not provide much room for the feedback or iteration to change the UI or design once the development starts. At the end of the phase, the requirements are generally frozen and goals are set. Waterfall model is not flexible for accommodating changes after this stage. So the user requirements that come from the UCD should be the final one, as any issue or feedback in the user input to accommodate the UI or any improvements cannot be easily incorporated. This derives that the whole UCD process collecting requirements, user research analysis, prototyping and user evaluation of the prototypes should also be completed within the Requirement phase.

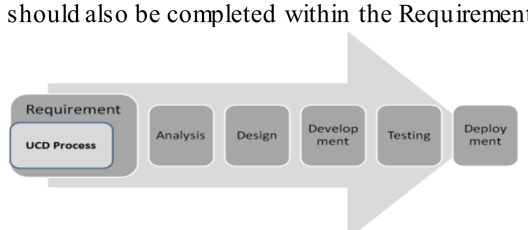

Figure 3 UCD Integrated approach for Waterfall model

B. Incremental (Iterative) Model

Incremental model [4] is building the application in small iterations, adding to the features in the product. This involves design, development and testing at each increment to ensure that the previous support is not broken along with new requirements. This is a predecessor of the Agile model which is one of the most common model used in the industry.

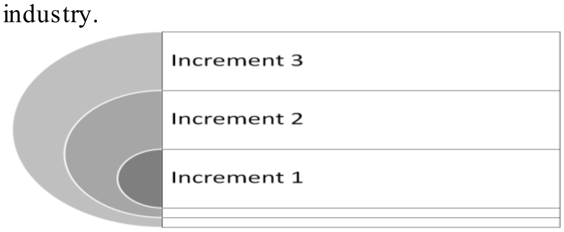

Figure 4 Incremental Model

Integrating the UCD into the iterative model is easier than the water fall model. In the iterative model the design, development and testing happen at the end of iteration; this gives scope to ensure that the user focus is taken care of for the deliverable of the iteration, providing better coverage. Also this model enables the feedback to be incorporated in the subsequent iterations. This results that the complete UCD process be implemented at each increment, for the deliverables associated to that stage. It is to be taken care that UCD approach should be restricted to take care of only the deliverables of that increment instead of planning the UI for complete software as it would happen in Waterfall model.

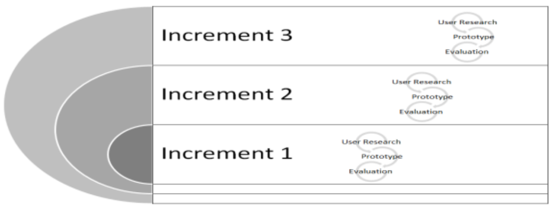

Figure 5 UCD Integrated approach for Incremental model

C. *Prototyping Model:*

There are SDLC models based on prototyping. This involves developing the prototypes and using them to get the feedback from the customer. Rapid prototyping (or Throw away prototyping), Evolutionary prototyping, etc are the variants of this model.

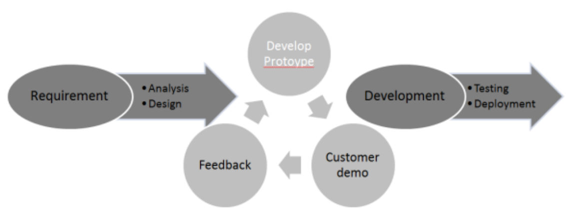

Figure 6 Prototyping Model

These models are effective to build upon the customer input and come up with the prototypes to discuss with the end user and finalize the requirements. This approach by itself is user oriented and by the very basic nature, UCD process. This requires the User Experience aspect to be integrated at each stage along with functional requirements, to ensure complete UCD perspective is met.

D. *Agile Model:*

Agile methodology (including XP and SCRUM which confirms to the Agile Manifesto [5]) aims at delivering small but completely functional features to the customer more frequently (Sprints).

The major reason that Agile is very successful with the developers and customer is that it allows the customers to understand what to expect and developers also get to know more on the deliverables and feedback. This is in contrast to the Waterfall model where the customer does not see the end product till the project is completed.

This way – agile model keeps the customer more engaged with the project development and also allows the feedback from customers to be incorporated earlier than the other model.

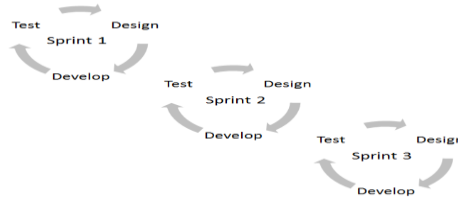

Figure 7 Agile Model

At first glance, UCD which aims to deliver a complete UI design for the developers to pick on does not seem to fit into the pieces of UI delivered to the customer in the Agile model. However an attempt on this path [9-16] has shown that this is feasible and more successful than first considered. The UCD can be incorporated into the Agile process more easily as both of them rely on the feedback from the customer about the frequent deliverables.

The approach is included below is derived from the study of multiple suggestions [9-16]. One of the most successful ways of this incorporation is from the references [17] explains how the study of Agile UCD integration show that the generic process with a sprint 0 for UCD to start and give the first deliverable that is supposed go out as a part of the formal Sprint 1 to the customer.

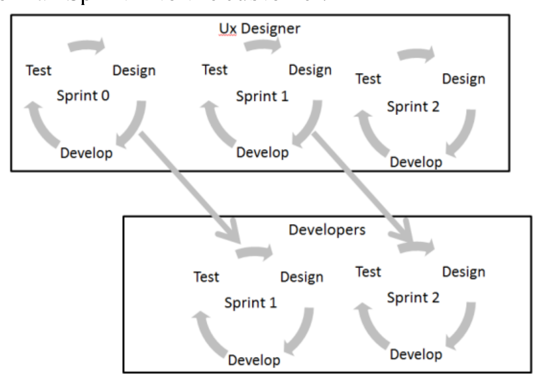

Figure 8 UCD Integrated Agile Model

VI. CONCLUSION

The paper proposes the integration approach for the different UCD in software lifecycle models. The list of models is exhaustive and the paper addresses the basic models that have been the forbearers for the more recent ones. The next stage of the research is to validate the proposed models in real life projects and measure the success in comparison to the previous models used. This study aims to become the basis on which future research can be done to identify a more detailed approach for integration of each of the models. The consolidation in this paper also gives path to explore the role and integration of UCD for other approaches to development like Aspect Driven, Model Driven and Test Driven Developments.

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